1. Agnostic learning

1. Risk and best-in-class target

For binary zero-one loss and $S = ((x_i, y_i))_{i=1}^m \sim D^m$,

$$R(h) = P_{(X,Y)\sim D}(h(X) \neq Y), \quad \hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^m \mathbf{1}_{\{h(x_i)\neq y_i\}}.$$

$\mathcal{H}$ is **agnostic PAC learnable** if there are one learner A and a sample complexity $m_{\mathcal{H}}$ such that, for every $\varepsilon, \delta \in (0,1)$, every joint distribution D, and every $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$P\Big(R(A(S)) \leq \inf_{h\in\mathcal{H}} R(h) + \varepsilon\Big) \geq 1 - \delta.$$

ε bounds **excess risk** above the best risk in the declared class; it is not distance from zero. The infimum need not be attained: compare with h_ρ satisfying $R(h_\rho) \leq \inf_h R(h) + \rho$, then let $\rho \to 0$.

2. Approximate ERM and tolerance roles

$h_S = A(S)$ is an η-approximate ERM output if

$$\hat{R}_S(h_S) \leq \inf_{h\in\mathcal{H}} \hat{R}_S(h) + \eta.$$

- η: empirical optimization error;
- ρ: proof-only population-comparator tolerance;
- α: empirical/population deviation tolerance;
- ε: requested final excess-risk tolerance.

Do not merge η with ρ. A successful output on one observed sample is not a PAC proof; the probability guarantee must hold over fresh samples for every allowed D.

3. Uniform convergence and ERM chain

The **uniform convergence** event is

$$E_\alpha = \Big\{\sup_{h\in\mathcal{H}}\big|\hat{R}_S(h) - R(h)\big| \leq \alpha\Big\}.$$

Uniform convergence requires $P(E_\alpha) \geq 1 - \delta$ for all distributions once m is sufficiently large. The shared event covers the data-selected h_S and the comparator h_ρ simultaneously. On E_α,

$$\begin{aligned} R(h_S) &\leq \hat{R}_S(h_S) + \alpha \\ &\leq \hat{R}_S(h_\rho) + \alpha + \eta \\ &\leq R(h_\rho) + 2\alpha + \eta \\ &\leq \inf_{h\in\mathcal{H}} R(h) + \rho + 2\alpha + \eta. \end{aligned}$$

Letting $\rho \to 0$ gives excess risk at most $2\alpha + \eta$. For exact ERM, $\eta = 0$ and the usual choice is $\alpha = \frac{\varepsilon}{2}$. Pointwise concentration for each fixed h does not by itself create this simultaneous event.

4. Finite class check

For fixed h under zero-one loss, Hoeffding gives

$$P\Big(\big|\hat{R}_S(h) - R(h)\big| > \alpha\Big) \leq 2\exp(-2m\alpha^2).$$

If $\mathcal{H}$ is finite, a union bound makes it sufficient that

$$m \geq \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\alpha^2}.$$

For exact agnostic ERM, substitute $\alpha = \frac{\varepsilon}{2}$:

$$m \geq 2\frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{\varepsilon^2}.$$

This has logarithmic dependence on class size and the agnostic $\frac{1}{\varepsilon^2}$ scale. It is not the realizable, consistent $\frac{1}{\varepsilon}$ route.

2. Learnability boundary

1. Fundamental Theorem

In **distribution-free binary classification with zero-one loss**, i.i.d. samples, suitable measurability, the following are equivalent:

1. uniform convergence;
2. every ERM rule is agnostic PAC;
3. agnostic PAC learnability;
4. realizable PAC learnability;
5. every ERM rule is realizably PAC;
6. $\mathrm{VCdim}(\mathcal{H}) < \infty$.

The equivalence is scoped: do not export it unchanged to arbitrary losses or nonmeasurable classes. The ERM chain gives uniform convergence → agnostic ERM success; agnostic learning implies realizable learning by restriction; No-Free-Lunch supplies the necessity of finite VC dimension.

2. No-Free-Lunch mechanism

For every binary learner A and sample size m with $2m \leq |\mathcal{X}|$, there is a realizable distribution with

$$P\Big(R(A(S)) \geq \frac{1}{8}\Big) \geq \frac{1}{7}.$$

For an infinite-VC class, choose a shattered support $C = \{x_1, \dots, x_{2m}\}$. A sample of size m sees at most m distinct points, so unseen points carry at least half the probability mass. Shattering realizes every target labeling; under random target bits, each unseen label remains a fair bit and any prediction is wrong with conditional probability $\frac{1}{2}$. Averaging identifies one fixed realizable hard target.

Quantifiers: for every learner, there exists a hard problem. Infinite class cardinality alone does not imply failure; the relevant richness is infinite VC dimension.

3. Random-sign complexity

1. Loss class and Rademacher complexity

Let $\mathcal{Z} = \mathcal{X} \times \{0,1\}$ and

$$g_h(x,y) = \mathbf{1}_{\{h(x)\neq y\}}, \quad \mathcal{G} = \{g_h : h \in \mathcal{H}\}.$$

For fixed $S = (z_1, \dots, z_m)$, the **empirical Rademacher complexity** is

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m \sigma_i g(z_i)\right],$$

where independent σ_i are uniform on $\{-1, +1\}$ and independent of S. Fix S, draw σ, maximize separately for that sign vector, then average. The distribution-averaged version is

$$\mathfrak{R}_m(\mathcal{G}) = \mathbb{E}_{S\sim D^m}\big[\hat{\mathfrak{R}}_S(\mathcal{G})\big].$$

$\mathcal{H}$ contains predictors; $\mathcal{G}$ contains their loss functions.

2. Ghost sample and symmetrization

A **ghost sample** $S' = (z_{1'}, \dots, z_{m'}) \sim D^m$ is independent of S and is used only inside the proof. It is not validation data or extra data given to the learner. With $\hat{E}_S[g] = m^{-1} \sum_i g(z_i)$,

$$\begin{aligned}
&\mathbb{E}_S \sup_g \left(\mathbb{E}_D[g] - \hat{E}_S[g]\right) \\
&\le \mathbb{E}_{S,S'} \sup_g \left(\hat{E}_{S'}[g] - \hat{E}_S[g]\right) \\
&= \mathbb{E}_{S,S',\sigma} \sup_g \frac{1}{m} \sum_i \sigma_i (g(z_{i'}) - g(z_i)) \\
&\le 2\mathfrak{R}_m(\mathcal{G}).
\end{aligned}$$

Factor ledger: $1, 1, 2$. The first step uses $\sup_g E[X_g] \le E[\sup_g X_g]$; exchangeable pairs license the signed equality; only supremum subadditivity creates the factor two.

3. Rademacher generalization bounds

For measurable $\mathcal{G}$ mapping $\mathcal{Z}$ to $[0, 1]$, with probability at least $1 - \delta$, simultaneously for every $g \in \mathcal{G}$,

$$\mathbb{E}_D[g] \le \hat{E}_S[g] + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\ln(\frac{1}{\delta})}{2m}}.$$

A valid data-dependent form is

$$\mathbb{E}_D[g] \le \hat{E}_S[g] + 2\hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\ln(\frac{2}{\delta})}{2m}}.$$

Apply the one-sided theorem to $\mathcal{G}$ and to $1 - \mathcal{G} = \{1 - g : g \in \mathcal{G}\}$, then split the confidence budget between the two directions to control absolute deviations. Constants depend on the one- or two-sided form. The ghost sample makes a paired empirical process; it is not needed because the unknown numerical value of $R(h)$ is somehow illegal in a probability bound.

4. Finite VC closes the arrow

For the zero-one loss class above and fixed labeled sample $S = (z_1, \dots, z_m)$, let

$$W_S = \{(g_h(z_1), \dots, g_h(z_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m.$$

For fixed labels, the prediction-to-loss map is coordinatewise and one-to-one, so $|W_S| \le \Pi_{\mathcal{H}}(m)$ and $\|w\|_2 \le \sqrt{m}$ for every $w \in W_S$. Massart's lemma gives

$$\hat{\mathfrak{R}}_S(\mathcal{G}) \le \sqrt{2\frac{\ln(|W_S|)}{m}} \le \sqrt{2\frac{\ln(\Pi_{\mathcal{H}}(m))}{m}}.$$

If $d = \mathrm{VCdim}(\mathcal{H})$ and $1 \le d \le m$, Sauer-Shelah gives

$$\Pi_{\mathcal{H}}(m) \le \sum_{j=0}^{d} \binom{m}{j} \le \left(e\frac{m}{d}\right)^d.$$

If $d = 0$, handle the boundary separately: $\Pi_{\mathcal{H}}(m) = 1$, so this finite-class Rademacher term is zero.

For $1 \le d \le m$, therefore, with probability at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}} \left| R(h) - \hat{R}_S(h) \right| = O\left(\sqrt{\frac{d \ln(e\frac{m}{d}) + \ln(\frac{1}{\delta})}{m}}\right).$$

Causal order: Sauer count → Massart random-sign bound → symmetrization/concentration → uniform convergence.

4. Rates and model choice

1. VC sample-complexity regimes

For finite VC dimension d:

- **Realizable PAC:** lower order $\Omega\left(\frac{d + \ln(\frac{1}{\delta})}{\varepsilon}\right)$; a standard upper bound is $O\left(\frac{d \ln(\frac{1}{\varepsilon}) + \ln(\frac{1}{\delta})}{\varepsilon}\right)$.
- **Agnostic PAC / uniform convergence:** $\Theta\left(\frac{d + \ln(\frac{1}{\delta})}{\varepsilon^2}\right)$ up to absolute constants.

Halving ε costs roughly $2\times$ samples in the realizable regime (plus logs), but $4\times$ in the agnostic regime. Realizability permits consistency detection; noisy agnostic learning must estimate differences between nonzero risks.

2. Structural risk minimization (SRM)

Let $\mathcal{H} = \cup_{k \in \mathbb{N}} \mathcal{H}_k$. Give every active class a weight $w_k > 0$ with $\sum_k w_k = 1$, set $\delta_k = w_k \delta$, and use a justified uniform radius $\varepsilon_k(m, \delta_k)$. Then, with probability at least $1 - \delta$, all active class certificates hold simultaneously:

$$R(h) \le \hat{R}_S(h) + \varepsilon_k(m, w_k\delta), \quad h \in \mathcal{H}_k.$$

SRM minimizes the right side across both h and k. A zero-weight class is excluded (equivalently, has infinite penalty). If a hypothesis lies in several classes, use its smallest valid assigned bound.

The family, confidence allocation, and theorem supporting each penalty must be declared; an attractive empirical-risk-plus-square-root expression alone is not the stated distribution-free certificate.

3. Select the proof route

- Fixed finite class, uniform upper bound: Hoeffding + union bound; decisive quantity $\ln|\mathcal{H}|$.
- Binary finite-VC class: Sauer/growth + Massart; decisive quantities $\Pi_{\mathcal{H}}(m)$ and d.
- Real-valued loss or useful sample geometry: ghost sample + Rademacher symmetrization; decisive quantity $\hat{\mathfrak{R}}_S(\mathcal{G})$.
- Overstrong universal learner/rate claim: shattered-set indistinguishable labels for a lower bound.

Use an upper-bound method for a generalization certificate and a shattered-set construction for impossibility. Infinite cardinality alone does not choose the route.